HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY AND EDUCATION | Master of Education Thesis
VIETJET AVIATION ACADEMY (VJAA) | VJC-VJAA-M-125

SAMPLE SCENARIO EXERCISE SYSTEM

COURSE: COMMUNICATION WITH FLIGHT CREW

Integrated Application of Large Language Models (LLMs)

5 Chapters | 10 Sample Scenarios | Scoring Rubrics | LLM Prompts

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DOCUMENT STRUCTURE

- Chapter 1: Aircraft Pushback & Towing (2 scenarios)
- Chapter 2: Communication Forms (2 scenarios)
- Chapter 3: Ground Handling Procedures (2 scenarios)
- Chapter 4: Ramp Safety (2 scenarios)
- Chapter 5: Signals & Prowords (2 scenarios)
- Appendix: LLM Usage Guide

EACH SCENARIO INCLUDES

+ Scenario context & situation details
+ Student task requirements (roles, actions)
+ ICAO-standard dialogue script
+ Suggested answer with explanation
+ Scoring rubric (5 criteria x 4 levels)
+ LLM prompt for secondary scenario generation

CHAPTER 1: AIRCRAFT PUSHBACK & TOWING OPERATIONS

This chapter covers: Pushback / Towing procedures; roles and responsibilities of the headset operator, flight crew, and tow vehicle driver; hand signals; and abnormal situation handling per VJC-VJAA-M-125 and VJC-GOP-M-01.

Chapter 1 | Scenario 1

Normal A320 Pushback - Conventional Towbar Tug

Difficulty: **Basic**

>> SCENARIO INFORMATION

Aircraft / Flight	VN-A649 (A320neo) - Flight VJ100
Location	Tan Son Nhat Airport (SGN) - Stand E05
Equipment	Conventional Towbar Tug - connected to nose gear
Conditions	Daytime, good visibility, all ground services complete
Pushback Clearance	Cleared by ATC - facing North
Learning Objective	Perform the full pushback communication procedure from start to disconnect

? STUDENT REQUIREMENTS

- Perform the complete communication sequence between the headset operator and flight crew, from start to disconnect.
- Identify the corresponding hand signals at each key step.
- List at least 5 actions PROHIBITED during the pushback procedure (per VJC regulations).

OK SUGGESTED ANSWER

A - ICAO-Standard Dialogue Script

Speaker	Communication Content (ICAO Standard)
Crew	Cockpit to Ground
Headset Man	Ground to Cockpit - Ready to pushback
Crew	Clear to pushback facing North
Headset Man	Pushback facing North. Release parking brakes.
Crew	Brakes released

Speaker	Communication Content (ICAO Standard)
Headset Man	Commencing pushback. Clear to start Engine No.2
Crew	Starting Engine No.2
Headset Man	N1 rotation - Engine No.2
Crew	Check
Crew	Starting Engine No.1
Headset Man	Clear to start Engine No.1
Headset Man	N1 rotation - Engine No.1
Crew	Check
Headset Man (tug stopped)	Pushback complete. Set parking brakes.
Crew	Brakes set
Crew	Start complete. Disconnect. Hand signals on right side.
Headset Man	Disconnecting. Hand signals on right side.
Headset Man	All clear. Goodbye.

B - Hand Signals at Key Steps

Step	Hand Signal Description
Parking brake set	Right fist raised overhead
Brakes released	Open fist (releasing gesture)
Ready to push	Thumbs up
Direction - North	Index finger points to nose, then straight ahead to the North
Set parking brake after pushback complete	Closed fist raised overhead again
Approved for disconnect	Wave hand away from aircraft (right side)

C - Prohibited Actions During Pushback (VJC Regulations)

- DO NOT rest feet on the brake pedals during pushback.
- DO NOT test external lighting from outside the flight deck at night.
- DO NOT allow the tug driver to brake suddenly, accelerate, or steer abruptly.
- DO NOT walk in front of or block the tug vehicle path.

- DO NOT maintain less than 3 m clearance from the towbar and nose gear while the aircraft is moving.
- DO NOT disconnect without receiving the order from the flight crew.

>> SCORING RUBRIC - Scenario 1.1 (Total: 20 pts)

Assessment Criteria	Max Score	1 - Below Standard	2 - Minimum	3 - Meets Standard	4 - Outstanding
[K] Understanding pushback procedure	**4**	Confuses multiple steps; cannot identify communication structure	Knows the procedure but fragmented; needs prompting to perform correctly	Correctly applies the procedure in familiar contexts	Demonstrates deep understanding; can explain to others and adapt flexibly
[S1] Use of ICAO standard terminology	**5**	Many terminology errors; does not use ICAO structure	Partly correct terminology; structural errors remain	Correct terminology and clear structure; minor errors only	Fully accurate and natural; flexible across all situations
[S2] Identification of correct hand signals	**4**	Cannot identify signals or makes serious errors	Identifies 1-2 signals; many still incorrect	Correctly identifies >= 4/6 signals	Identifies and accurately describes all signals
[S3] Listing prohibited actions	**4**	Lists fewer than 2 prohibited actions	Lists 2-3 prohibited actions	Lists 4-5 with brief explanation	Lists >= 5 with safety rationale for each
[A] Professional attitude & discipline	**3**	No compliance awareness; skips procedures	Complies only when prompted	Proactively complies with all steps	Exemplary; reminds teammates when needed
	20	< 60% = Below Standard		60-79% = Meets Standard	>= 80% = Outstanding

AI SECONDARY SCENARIO GENERATION PROMPT

SECONDARY SCENARIO GENERATION PROMPT (LLMs: Claude / GPT-4 / Gemini)

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ROLE: You are an aviation ground operations training expert at VJAA (Vietjet
Aviation Academy). REFERENCE DOCUMENT: VJC-VJAA-M-125 "Communication with
Flight Crew" - Chapter 1: Aircraft Pushback & Towing. BASE SCENARIO: Normal
A320 pushback at TSN, conventional towbar tug, facing North, daytime, normal
conditions.  TASK: Create 2 NEW secondary scenarios based on the above:
[SCENARIO A - INTERMEDIATE level] - Changes: Aircraft A321, Noi Bai Airport
(HAN), TOWBARLESS tug (no towbar) - Add: APU INOP - GPU required for engine
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start - Student requirement: Full communication procedure + explain
differences when using TOWBARLESS tug  [SCENARIO B - ADVANCED level] -
Changes: Aircraft A330, Phu Quoc Airport (PQC), nighttime (23:00) - Add:
Headset fails mid-operation; must switch to hand signals - Add: Wing walker
signals stop due to obstacle on right side - Student requirement: Handle
sequentially; write full dialogue + stop/continue decision  OUTPUT FORMAT PER
SCENARIO: 1. Scenario information table (Aircraft / Airport / Equipment /
Conditions) 2. Student requirements (3-5 bullets) 3. Dialogue script table
(Speaker | ICAO-standard content) 4. Suggested answer with explanation 5.
Scoring rubric: 5 criteria x 4 levels (total 20 pts) 6. 2 open discussion
questions  CONSTRAINTS: Use ICAO standard English for all radio calls; do not
invent procedures not in VJC-VJAA-M-125.
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Chapter 1 | Scenario 2

Emergency: Towbar Pin Shear - Safety Pin Detached During Pushback

Difficulty **Advanced**

>> SCENARIO INFORMATION

Aircraft / Flight	VN-A645 (A320) - Pushback in progress at Stand F03
Location	Noi Bai Airport (HAN)
Incident	Towbar safety pin sheared - towbar head DETACHED from nose gear
Timing	Aircraft moving slowly; engines NOT yet started
Reference Document	Section 1.2.3 - VJC-GOP-M-01 (Emergency Response Plan - ERP)
Learning Objective	Identify the incident, act in correct priority order, and complete post-incident reporting

? STUDENT REQUIREMENTS

- Identify ALL emergency actions in the correct priority order (per Section 1.2.3).
- Accurately describe the hand signal for an emergency stop.
- Outline all post-incident reporting steps per VJC procedures.
- Explain: Why must the scene be preserved, and who has authority to authorize removal of equipment?

OK SUGGESTED ANSWER

A - Emergency Actions in Priority Order

Priority	Responsible	Immediate Action	Reference
1 - Immediate	Tug Driver	Declare incident - apply emergency brakes on tug	Towbar head detached detected
2 - Immediate	Headset Operator	Give EMERGENCY STOP signal: both arms thrust upward, wands crossed in X shape - fast and decisive	ICAO Annex 2
3 - Immediate	Headset Operator	Radio: "STOP! STOP! Towbar failure!" - crew applies brakes immediately	Section 1.2.3
4 - After tug stops	Tug Driver	Safely move tug clear of aircraft	Equipment safety
5 - Immediately after Step 4	Headset Operator	Chock nose gear immediately	Section 1.2.3
6 - Simultaneously	All	CEASE ALL ground service activity	Overall safety
7 - Next	Headset Operator	Immediately notify CRS/Engineering, VJC Representative, Airport Authority	Incident reporting
8 - Ongoing	All	PRESERVE scene - do not move any equipment before investigation	Section 1.2.3

B - Emergency Stop vs Normal Stop Hand Signal

Criteria	Normal Stop	EMERGENCY Stop
Execution	Both arms slowly raised overhead until wands meet	Both arms SUDDENLY thrust upward - wands cross in X shape - fast and forceful
Speed	Slow, controlled	Sudden, powerful, no hesitation
Context	Aircraft reaches planned stop line	Unexpected hazard detected
Additional signal	Not required	Combined with radio/headset alert

C - Post-Incident Reporting Procedure

- Immediately notify VJC Ground Operations Representative, Airport Authority, and flight crew.
- Preserve the scene and await CRS/Engineering inspection.
- Complete an incident report: time, location, sequence of events, witnesses, estimated damage.
- Photograph the scene from multiple angles: wide shot, tug position, detached towbar, nose gear.
- Submit internal incident report and cooperate with investigation per ERP (VJC-GOP-M-01).

>> SCORING RUBRIC - Scenario 1.2 (Total: 25 pts)

Assessment Criteria	Max Score	1 - Below Standard	2 - Minimum	3 - Meets Standard	4 - Outstanding
[K1] Incident recognition & classification	**4**	Does not recognize this as an emergency	Recognizes incident but misclassifies severity	Correctly identifies and names the incident type	Instantly identifies; clearly analyzes cause and danger level
[S1] Actions in priority order	**6**	Completely wrong order or omits critical steps	Correct order for 3-4 steps; still wrong or missing others	Correct order for >= 6/8 steps; can explain reasoning	Fully correct order; fast reflex; no prompting needed
[S2] Emergency stop hand signal	**5**	Does not know signal or confuses it with normal stop	Describes signal but wrong on speed/force	Correctly describes signal and distinguishes from normal stop	Accurate description and correctly demonstrates both signal types
[S3] Post-incident reporting procedure	**5**	Unaware of reporting procedure or skips it entirely	Names 1-2 reporting steps	Names 4-5 reporting steps with explanation	Complete, correct sequence with specific document references
[A] Safety compliance - scene preservation	**5**	Suggests moving equipment immediately; does not preserve scene	Knows scene must be preserved but cannot explain why	Correctly preserves scene and explains reason for investigation	Performs correctly and knows exactly who has authority to authorize equipment removal
	25	< 60% = Below Standard		60-79% = Meets Standard	>= 80% = Outstanding

AI SECONDARY SCENARIO GENERATION PROMPT

SECONDARY SCENARIO GENERATION PROMPT (LLMs: Claude / GPT-4 / Gemini)

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ROLE: You are an aviation safety expert at VJAA. REFERENCE DOCUMENT: VJC-VJAA-M-125 Chapter 1 (Sections 1.2.2, 1.2.7, 1.2.11) and VJC-GOP-M-01 (ERP). BASE SCENARIO: Towbar safety pin sheared and towbar head detached from A320 nose gear at HAN.  TASK: Create 1 secondary scenario - choose RANDOMLY from the following incident types: (a) Safety pin sheared but towbar NOT detached (Section 1.2.2) (b) Engine fire during pushback (Section 1.2.7) - ADVANCED
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level (c) Headset completely loses connection mid-operation (Section 1.2.11)
(d) Wing walker signals STOP due to obstacle on the right wing clearance zone
OUTPUT FORMAT: 1. Scenario information table 2. Incident timeline (>= 5
milestones) 3. Student analysis tasks: (a) Recognition, (b) Priority actions,
(c) Communication, (d) Reporting 4. Complete suggested answer 5. Rubric: 5
criteria x 4 levels (total 25 pts) 6. 3 critical-thinking discussion questions
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CHAPTER 2: FORMS OF CREW COMMUNICATION

This chapter covers: Arrival communication; Departure communication; Use of GPU, ACU, ASU; Communication for COMAC C909 aircraft; Handling technical abnormalities during communication.

Chapter 2 | Scenario 1

Arrival: BEACON Still Illuminated + Connect GPU & ACU

Difficulty: **Intermediate**

>> SCENARIO INFORMATION

Aircraft / Flight	VN-A632 (A320) - Just taxied into Stand B12
Location	Da Nang Airport (DAD)
Observed Conditions	Nose gear chocked - BEACON still illuminated - Crew requests GPU & ACU
Abnormality	Anti-collision BEACON light still flashing after aircraft is on stand
Reference Document	Section 2.2 - VJC-VJAA-M-125 (Normal Arrival)
Learning Objective	Correctly perform 3-step procedure: establish comms -> handle BEACON -> connect GPU/ACU

? STUDENT REQUIREMENTS

- Perform the full 3-step communication procedure: (1) Establish contact, (2) Handle BEACON, (3) Connect GPU & ACU.
- Explain WHY the BEACON must be addressed before allowing ground staff to approach the aircraft.
- State what happens if ground staff approach the aircraft while the BEACON is still illuminated.

OK SUGGESTED ANSWER

Step 1 - Establish Contact & Confirm Chocks In

Speaker	Communication Content (ICAO Standard)
Headset Man	Ground to Cockpit
Crew	Cockpit to Ground - Go ahead
Headset Man	Chocks in place
Crew	Brakes released

Step 2 - Handle BEACON Still Illuminated

Speaker	Communication Content (ICAO Standard)
Headset Man	Ground to Cockpit - BEACON lights still ON
Crew	Roger. BEACON set OFF
Headset Man	BEACON OFF confirmed. Thank you.

Step 3 - Connect GPU & ACU

Speaker	Communication Content (ICAO Standard)
Headset Man	The External Ground Power Unit is available
Crew	Supply External Ground Power
Headset Man	External Ground Power is available - connected
Headset Man	The External Air Conditioning Unit is available
Crew	Supply External Air Conditioning Flow
Headset Man	External Air Conditioning Flow is available - connected

Explanation: Why Handle BEACON First

The BEACON (flashing red anti-collision light) signals that the aircraft may move or engines may be activated at any moment. While the BEACON is ON, ground personnel MUST NOT approach hazardous zones (engines, wings, landing gear). Confirming BEACON OFF is the mandatory safety prerequisite before any ground staff can safely approach.

>> SCORING RUBRIC - Scenario 2.1 (Total: 20 pts)

Assessment Criteria	Max Score	1 - Below Standard	2 - Minimum	3 - Meets Standard	4 - Outstanding
[K1] Understanding 3-step procedure	**4**	Does not know the sequence or skips the BEACON step	Knows steps but performs them out of order	Performs all 3 steps in correct order without omission	Performs correctly and explains the rationale for each step
[S1] ICAO communication terminology	**5**	Many terminology errors; uses everyday language instead	50-70% correct; missing key standard phrases	>= 80% correct terminology; clear structure	Fully accurate and natural; no prompting needed

Assessment Criteria	Max Score	1 - Below Standard	2 - Minimum	3 - Meets Standard	4 - Outstanding
[K2] Explanation of BEACON handling	**4**	Cannot explain or provides incorrect explanation	Vague explanation; does not identify specific hazard	Correctly explains the hazard and relevant safety rule	Deep explanation with a specific example and document reference
[S2] GPU/ACU communication handling	**4**	Does not know GPU/ACU phrases or skips this step	Attempts but phrases are not standard	Correctly uses standard GPU and ACU phrases	Correct and can also handle when either unit is not available
[A] Safety: No approach while BEACON is ON	**3**	Unaware of the rule; may approach while BEACON is on	Knows the rule but slow to react	Proactively halts all activity until BEACON confirmed OFF	Immediate reflex; alerts surrounding personnel
	20	< 60% = Below Standard		60-79% = Meets Standard	>= 80% = Outstanding

AI SECONDARY SCENARIO GENERATION PROMPT

SECONDARY SCENARIO GENERATION PROMPT (LLMs: Claude / GPT-4 / Gemini)

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ROLE: You are a "Communication with Flight Crew" instructor at VJAA. REFERENCE
DOCUMENT: VJC-VJAA-M-125 - Section 2.2 (Arrival). BASE SCENARIO: Arrival with
BEACON still ON + GPU/ACU request - A320 at DAD.  TASK: Create 2 secondary
scenarios:  [SCENARIO A - INTERMEDIATE level] - Normal arrival BUT: crew does
not respond to first radio call - Add: Jetway not yet retracted when headset
man checks - Student task: Handle in correct order - call crew again +
coordinate jetway retraction + BEACON communication  [SCENARIO B - ADVANCED
level] - Nighttime arrival with light fog (visibility ~500m) - APU INOP
immediately after chocking -> GPU required urgently - Simultaneously:
passengers still on board; refueling in progress - Student task: Full
communication procedure + special safety precautions for refueling with
passengers on board  OUTPUT FORMAT: Info table - Student tasks - Dialogue
script - Answer - 20-pt rubric - 2 discussion questions
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Chapter 2 | Scenario 2

Departure - COMAC C909: Towbar Failure + Engine Start via GPU/ASU with APU INOP

Difficulty **Advanced**

>> SCENARIO INFORMATION

Aircraft / Flight	COMAC C909 - Pushback at Phu Quoc Airport (PQC)

Incident	Towbar malfunction mid-pushback - operation stopped; towbar being replaced
Special Condition	APU INOP - GPU + ASU required to start Engine No.2 first
Engine Start Sequence	Engine No.2 first, then No.1 (differs from standard Airbus procedure)
Reference Document	Sections 2.3.3 and 2.3.4 - VJC-VJAA-M-125 (COMAC C909)
Learning Objective	Handle towbar failure + engine start via GPU/ASU + compare Airbus vs COMAC communication

? STUDENT REQUIREMENTS

- Perform the full communication procedure: (1) Handle towbar failure, (2) Engine start via GPU/ASU after pushback complete.
- Compare communication phrase differences between Airbus A320/A330 and COMAC C909 (at least 3 points).
- Explain why Engine No.2 must be started first when APU is INOP.

OK SUGGESTED ANSWER

Part A - Towbar Failure Handling (Section 2.3.4)

Speaker	Communication Content (ICAO Standard)
Headset Man	Ground to Cockpit - Please set parking brake. Towbar problem.
Crew	Parking brake set. Standing by.
(After towbar replacement)	---
Headset Man	Ground to Cockpit - Towbar replaced and reconnected. Please release parking brake.
Crew	Brakes released. Ready to continue pushback.
Headset Man	Commencing pushback.

Part B - Engine Start via GPU/ASU with APU INOP

Speaker	Communication Content (ICAO Standard)
Headset Man	Pushback complete. Set parking brake.
Crew	Brakes set.

Speaker	Communication Content (ICAO Standard)
Headset Man	External bleed (ASU) connected. External power (GPU) connected. Clear to start Engine No.2.
Crew	Starting Engine No.2.
Headset Man	N1 rotation - Engine No.2.
Crew	Engine No.2 started. Please remove external ASU and GPU.
Headset Man	Roger. Disconnecting external bleed and power. Standby.
Headset Man	External bleed and power disconnected. Clear to start Engine No.1.
Crew	Starting Engine No.1. Engine No.1 started. Start complete. Disconnect headset.
Headset Man	Disconnecting. Hand signals on right side. All clear. Goodbye.

Part C - Airbus A320/A330 vs COMAC C909 Comparison

Situation	Airbus A320/A330	COMAC C909
Initial contact	Cockpit to Ground / Ground to Cockpit	Ground to Cockpit - Go ahead (crew more proactive)
Push direction	Crew reports: "Clear to pushback facing..."	Headset man asks: "Please confirm nose direction?"
Readiness confirmation	Immediately when crew gives clearance	Headset man confirms: "All set and ready"
Start with APU INOP	Start Eng No.2 first (right wing)	Start Eng No.2 first - different ASU/GPU phrase structure
Engine start sequence	No.2 then No.1	No.2 then No.1 (same) but confirmation phrases differ
Disconnect	Start complete, disconnect, hand signals on right	Same but with additional step confirming GPU/ASU offline

>> SCORING RUBRIC - Scenario 2.2 (Total: 25 pts)

Assessment Criteria	Max Score	1 - Below Standard	2 - Minimum	3 - Meets Standard	4 - Outstanding
[K1] Towbar failure procedure	**4**	Does not know whether to stop or correct sequence	Knows to stop but missing brake-set or standby step	Correct sequence: stop -> brake -> wait -> resume	Correct and smooth communication; no time wasted

Assessment Criteria	Max Score	1 - Below Standard	2 - Minimum	3 - Meets Standard	4 - Outstanding
[S1] GPU/ASU engine start communication (COMAC)	**6**	Does not know COMAC GPU/ASU phrases	Uses Airbus phrases instead; incorrect for COMAC procedure	Uses correct COMAC phrases in correct sequence	Fully accurate; can also handle when GPU or ASU is not yet ready
[K2] Airbus vs COMAC comparison	**6**	Cannot identify any differences	Identifies 1-2 correct differences	Identifies >= 3 differences with explanation	Identifies >= 5 differences with practical impact analysis
[S2] Explain why Engine No.2 starts first	**5**	Cannot explain or explanation is incorrect	Vague reference to engine position only	Correctly explains: Eng No.2 (right wing) is closest to ASU	Correct explanation + states consequences if sequence is reversed
[A] Discipline and accuracy in execution	**4**	Multiple uncorrected errors	Errors present but self-corrected after prompting	Few errors; self-corrects during execution	No errors; fast and professional
	25	< 60% = Below Standard		60-79% = Meets Standard	>= 80% = Outstanding

CHAPTER 3: GROUND HANDLING PROCEDURES

This chapter covers: Walk-around inspection; Refueling procedures (automatic and manual); CRS/GO authority; Turnaround procedures; Coordination with related departments.

Chapter 3 | Scenario 1

Walk-Around: Abnormality Found on Horizontal Stabilizer During Turnaround

Difficulty **Intermediate**

>> SCENARIO INFORMATION

Aircraft	VN-A651 (A320neo) - Turnaround at Can Tho Airport (VCA)
Timing	GO staff performing walk-around inspection
Finding	Significant unusual scratch on left horizontal stabilizer
Context	Inbound passengers have disembarked; outbound passengers preparing to board
Reference Document	Section 3.2 - VJC-VJAA-M-125 + VJC-GOP-M-01 (Damage Report)
Learning Objective	Detect walk-around abnormalities, respond correctly, and determine decision authority

? STUDENT REQUIREMENTS

- Describe the complete walk-around procedure and identify where the scratch was found.
- State the immediate actions upon discovering an abnormality on the aircraft structure.
- Identify: Who has the authority to determine whether the aircraft is airworthy for continued operation?
- Answer: While awaiting CRS assessment, may boarding continue?

OK SUGGESTED ANSWER

Part A - Immediate Actions on Discovering Abnormality

Step	Action	Reason
1	STOP walk-around immediately - do not continue to other areas	Prioritize handling the abnormality first
2	Photograph the scratch from multiple angles	Evidence for investigation and reporting

Step	Action	Reason
3	DO NOT allow boarding - notify boarding staff immediately	Passenger safety before aircraft is confirmed airworthy
4	Immediately notify VJC Representative (GO) and CRS/Engineering	This is CRS authority - not GO
5	Inform flight crew via headset of the abnormality finding	Crew must be aware to prepare for possible delay
6	Complete abnormality discovery report and await CRS assessment	Mandatory reporting procedure

Part B - Authority to Determine Airworthiness

Per VJC regulations: ONLY the licensed CRS (Certificate of Release to Service) engineer has the authority to assess damage severity and confirm the aircraft is airworthy. The GO CANNOT independently decide whether the aircraft can continue operations, regardless of how minor the abnormality appears.

Part C - Boarding Decision Answer

NO. While awaiting CRS assessment, boarding MUST NOT continue. This is a mandatory safety principle: an aircraft not yet confirmed airworthy by CRS must not proceed with any passenger-related activity.

>> SCORING RUBRIC - Scenario 3.1 (Total: 20 pts)

Assessment Criteria	Max Score	1 - Below Standard	2 - Minimum	3 - Meets Standard	4 - Outstanding
[K1] Walk-around procedure & abnormality detection	**4**	Does not know walk-around or omits critical steps	Knows procedure but cannot accurately locate the abnormality	Correctly performs walk-around and locates the scratch	Correctly performs and can analyze potential severity
[S1] Immediate actions (6 steps)	**5**	Allows boarding to continue or omits reporting step	Stops boarding but misses photographing or CRS notification	Performs >= 5/6 steps in correct order	Complete and fast; explains rationale for each step
[K2] CRS authority	**5**	Believes GO can decide independently	Knows CRS is needed but cannot explain why	Correctly identifies authority and explains reason	Deep explanation + states legal consequences if GO decides incorrectly
[S2] Boarding halt decision	**4**	Allows passengers to continue	Halts boarding but does not	Halts boarding and correctly notifies all	Correct action + professionally

Assessment Criteria	Max Score	1 - Below Standard	2 - Minimum	3 - Meets Standard	4 - Outstanding
		boarding while awaiting CRS	notify boarding staff	relevant parties	informs waiting passengers
[A] Safety procedure compliance	**2**	Tends to rush; bypasses procedures	Complies when reminded	Proactively complies without prompting	Exemplary; proactively reminds colleagues
	20	< 60% = Below Standard		60-79% = Meets Standard	>= 80% = Outstanding

CHAPTER 4: SAFETY IN AIRCRAFT GROUND HANDLING OPERATIONS

This chapter covers: Safety during aircraft movement on the ramp; Lightning SHUTDOWN procedures; Handling operations in adverse weather; GSE collision response; FOD (Foreign Object Damage) handling.

Chapter 4 | Scenario 1

Ramp SHUTDOWN Due to Lightning - Managing 3 Simultaneous Tasks

Difficulty **Advanced**

>> SCENARIO INFORMATION

Location	Tan Son Nhat Airport (SGN) - 14:30
SHUTDOWN Order	ATC/Ramp Control issues SHUTDOWN: lightning within 6 km of airport
Task 1 (in progress)	Pushback VN-A641 at Stand A06 - tug connected, NOT yet moving
Task 2 (in progress)	Monitoring refueling of VN-A632 at Stand B12 - refueling in progress
Task 3 (in progress)	Marshalling passengers boarding VN-A650 at Stand C03 via OUTDOOR stairs
Learning Objective	Correctly prioritize and execute appropriate actions for all 3 simultaneous tasks

? STUDENT REQUIREMENTS

- Determine the correct priority order for handling all 3 tasks upon receiving the SHUTDOWN order - EXPLAIN YOUR REASONING.
- Describe the specific action for each task in priority order.
- Answer: When can operations resume after SHUTDOWN? Who has the authority?

OK SUGGESTED ANSWER

Part A - Priority Order and Actions

Priority	Task	Immediate Action	Reason
#1 Most Urgent	Passengers boarding via outdoor stairs (C03)	STOP boarding immediately. Guide passengers inside the terminal via nearest route. DO NOT shelter under the aircraft or allow passengers to remain on the ramp.	People > Equipment. Passengers outdoors are the highest lightning risk.

Priority	Task	Immediate Action	Reason
#2 Important	Refueling VN-A632 (B12)	STOP refueling immediately. Remove headset (DO NOT wear during lightning). Brief radio call to crew: SHUTDOWN. Move to safe area.	Refueling + lightning = high fire/explosion risk. Headset = electrical conductor.
#3 Lower Risk	Pushback VN-A641 (A06) - not yet moving	Leave as-is (tug connected, not yet moving). Brief radio call to crew: STANDBY. Move to shelter position.	Aircraft not yet moving; risk is lower. Can be handled after others.

Part B - When Can Operations Resume

Operations may resume ONLY AFTER: (1) Ramp Control officially cancels the SHUTDOWN order, and (2) A formal all-clear notice is issued. The GO MUST NOT resume any activity until an official notification is received. If unclear: CALL Ramp Control to confirm.

>> SCORING RUBRIC - Scenario 4.1 (Total: 25 pts)

Assessment Criteria	Max Score	1 - Below Standard	2 - Minimum	3 - Meets Standard	4 - Outstanding
[K1] Understanding the SHUTDOWN order	**3**	Does not know what SHUTDOWN means; continues working	Knows to stop but unsure what to do next	Understands and knows to halt all relevant activity	Deeply understands: can distinguish SHUTDOWN from a routine weather advisory
[S1] Correct priority order	**7**	Completely wrong order (e.g., prioritizes equipment over people)	Correct priority for 1-2 of the 3 tasks	Correct priority for all 3; explains why passengers come first	Correct priority + clear risk analysis + contingency for each
[S2] Specific actions for each task	**8**	Critical actions missing for >= 2 tasks	Correct actions for 1-2 tasks; 1 still wrong	Correct actions for all 3; minimal omissions	Fully correct, fast, and can explain each decision
[K2] Conditions for resuming after SHUTDOWN	**4**	Resumes on own judgment when thunder sounds distant	Knows to wait but unclear who has authority	Correctly states: resume only on official Ramp Control notification	Correct + knows how to verify with Ramp Control if unclear
[A] Composure & team coordination under pressure	**3**	Panics; cannot coordinate	Calm but slow and disorganized	Calm; acts in orderly sequence	Calm, professional; coordinates the entire ground team

Assessment Criteria	Max Score	1 - Below Standard	2 - Minimum	3 - Meets Standard	4 - Outstanding
	25	< 60% = Below Standard		60-79% = Meets Standard	>= 80% = Outstanding

CHAPTER 5: AIRCRAFT MARSHALLING SIGNALS & COMMUNICATION PROWORDS

This chapter covers: Marshalling signals for guiding aircraft to stand; ICAO NATO Phonetic Alphabet; Standard phraseology (Prowords); Technical signals at the stand.

Chapter 5 | Scenario 1

Marshalling A330 Into Stand - Emergency Stop Required

Difficulty: **Intermediate**

>> SCENARIO INFORMATION

Aircraft	VN-A810 (A330) - Taxiing into Stand at Noi Bai Airport (HAN)
Scenario	Marshaller giving "straight ahead" signal. Aircraft 8 m from stop line.
Abnormality	Right wing walker signals alarm: baggage tug still parked in safety zone behind right wing
Reference Document	Section 5.2 - VJC-VJAA-M-125 (Marshalling Signals)
Learning Objective	Execute full marshalling signal sequence + distinguish normal stop from emergency stop

? STUDENT REQUIREMENTS

- Describe the COMPLETE marshalling signal sequence from aircraft approach to emergency stop.
- Accurately describe the emergency stop signal and distinguish it from the normal stop signal.
- Describe the actions to be taken after the aircraft has stopped safely.

OK SUGGESTED ANSWER

Part A - Marshalling Signal Sequence

Step	Signal Name	Execution Description
1	Stand identification	Raise both arms fully extended overhead, wands pointing up
2	Wing walkers in position	Right arm raised; left arm with wand pointing down at body
3	Straight ahead	Bend elbows; wands move up and down from chest to head (continuous)

Step	Signal Name	Execution Description
4	Slow down (near stop line)	Wands move in gentle "patting" motion from waist to knee level
5 EMERGENCY	EMERGENCY STOP	Sudden: BOTH arms thrust upward, wands cross in X shape - forceful and decisive

Part B - Normal Stop vs Emergency Stop

Criteria	Normal Stop	EMERGENCY Stop
Execution	Both arms slowly raised overhead until wands meet	Both arms SUDDENLY flung upward; wands cross in X - no hesitation
Speed	Slow, controlled	Extremely fast, sudden, forceful
Context	Aircraft reaches planned stop point	Unexpected hazard detected
Additional signal	Not required	Combined with radio/headset emergency call

Part C - Actions After Aircraft Stops

- Maintain "Hold position" signal (both arms down at 45 degrees) until aircraft is completely stationary.
- Order the baggage tug to clear the safety zone IMMEDIATELY.
- Once path is clear: resume guiding aircraft to stand or signal for marshaller handoff.
- Report the incident to the Ramp Supervisor after aircraft is safely on stand.

>> SCORING RUBRIC - Scenario 5.1 (Total: 20 pts)

Assessment Criteria	Max Score	1 - Below Standard	2 - Minimum	3 - Meets Standard	4 - Outstanding
[K] Understanding the marshalling signal system	**4**	Cannot name the basic signals	Names 2-3 signals but wrong sequence	Correctly names >= 4/5 signals in correct order	Correctly names all; accurately describes each motion
[S1] Emergency stop signal description	**5**	Does not know or confuses with normal stop	Correctly describes form but wrong on speed/force	Correctly describes and distinguishes from normal stop	Accurate description + explains the technical reason for the difference
[S2] Actions after stop	**4**	Does not know what to do after	Knows to clear the tug but	Performs 3-4 correct post-stop steps	Full, correct sequence

Assessment Criteria	Max Score	1 - Below Standard	2 - Minimum	3 - Meets Standard	4 - Outstanding
		emergency stop	omits reporting step		including incident report
[S3] Distinguish between 2 stop signals	**4**	Cannot distinguish	Distinguishes 1-2 criteria	Distinguishes all 4 criteria in the comparison	Distinguishes all + provides example situations for each
[A] Reflex & professional conduct	**3**	Slow to react when situation arises	Reacts but needs prompting	Fast reflex; correct immediate action	Outstanding reflex; leads the entire ground team
	20	< 60% = Below Standard		60-79% = Meets Standard	>= 80% = Outstanding

AI SECONDARY SCENARIO GENERATION PROMPT

SECONDARY SCENARIO GENERATION PROMPT (LLMs: Claude / GPT-4 / Gemini)

```
ROLE: You are a marshalling instructor at VJAA. REFERENCE DOCUMENT: VJC-VJAA-M-125 Section 5.2 (Marshalling Signals). BASE SCENARIO: Marshalling A330 into stand at HAN - emergency stop due to tug in safety zone.  TASK: Create 2 secondary scenarios:  [SCENARIO A - Nighttime] Guide A320 to stand at night (23:00) - marshaller using illuminated LED wands. Scenario: When aircraft is 15 m from stop line, ONE wand's light suddenly fails. Student tasks: (a) Identify handling options, (b) One-wand substitute signal, (c) Safety implications. Rubric: 20 pts.  [SCENARIO B - NATO Phonetic + Prowords] Headset operator needs to confirm callsign "VN-A649 VIETJET AIR" by radio. Multi-part exercise: (a) Full NATO Phonetic Alphabet pronunciation (complete answer key) (b) 5 callsign spelling exercises (VN-A6XX to A8XX) (c) One communication transcript WITH ERRORS (wrong Prowords/terminology) - students find errors and rewrite correctly. Provide complete answer key and classroom usage guide.
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Chapter 5 | Scenario 2

Prowords & NATO Phonetic: Handling Miscommunication During Pushback

Difficulty **Basic**

>> SCENARIO INFORMATION

Situation	A320 Pushback - Headset man transmits: "Pushback facing North. Release brakes."
Abnormality 1	Crew misunderstands pushback direction - asks: "Say again the heading after pushback?"

Abnormality 2	Crew confirms brakes released but headset man cannot hear clearly due to loud engine noise
Learning Objective	Use correct Prowords to resolve misunderstanding, re-confirm information, and ensure safe communication
Key Prowords	SAY AGAIN | CORRECTION | CONFIRM | ROGER | WORDS TWICE

? STUDENT REQUIREMENTS

- Use correct Prowords to handle the FULL misunderstanding situation above (rewrite complete dialogue).
- Explain the exact meaning and correct usage of all 5 key Prowords in the table.
- Spell using NATO Phonetic Alphabet: "NORTH", "VN-A649", "VIETJET".

OK SUGGESTED ANSWER

Part A - Proword Handling Script (ICAO Standard)

Speaker	Communication Content (ICAO Standard)
Headset Man	Pushback facing North. Release parking brakes.
Crew	Say again the heading after pushback?
Headset Man	Correction. I say again: Pushback facing NOVEMBER - OSCAR - ROMEO - TANGO - HOTEL. Heading NORTH. Confirm.
Crew	Affirm. Heading North confirmed. Brakes released.
Headset Man	(Engine noise - cannot hear clearly) Say again. Unable to confirm brake status. Words twice.
Crew	Brakes released. Brakes released.
Headset Man	Roger. Brakes released confirmed. Commencing pushback.

Part B - 5 Key Prowords Explained

Proword	Exact Meaning	When to Use	NOT used when
SAY AGAIN	Request to repeat all or part of the previous transmission	Did not hear clearly or need confirmation	Requesting new/additional information
CORRECTION	An error was made in a previous transmission - correct version follows	You said something wrong and need to correct it immediately	Correcting the other party's error

Proword	Exact Meaning	When to Use	NOT used when
CONFIRM	Request to verify specific information (clearance, heading, action)	Need the other party to verify important information	Simply expressing thanks
ROGER	All of your last transmission has been received (does NOT mean "I will comply")	Confirming you heard and understood	Confirming you will perform an action (use WILCO instead)
WORDS TWICE	Request each word/group to be spoken TWICE due to poor channel quality	Difficult communication conditions; high noise or interference	Normal communication conditions

Part C - NATO Phonetic Alphabet Pronunciation

Text to Spell	NATO Phonetic Pronunciation	Notes
NORTH	NOVEMBER - OSCAR - ROMEO - TANGO - HOTEL	5 letters
VN-A649	VICTOR - NOVEMBER - Alfa - SIX - FOUR - NINER	Numbers: SIX, FOUR, NINER (not six/four/nine)
VIETJET	VICTOR - INDIA - ECHO - TANGO - JULIETT - ECHO - TANGO	7 letters

>> SCORING RUBRIC - Scenario 5.2 (Total: 20 pts)

Assessment Criteria	Max Score	1 - Below Standard	2 - Minimum	3 - Meets Standard	4 - Outstanding
[S1] Rewrite dialogue using correct Prowords	**7**	No Prowords used or all used incorrectly	2-3 Prowords used correctly; context errors remain	>= 4/5 Prowords used in correct context	All used correctly and naturally; dialogue flows smoothly
[K1] Explain 5 Prowords	**5**	Incorrect explanation for >= 3 Prowords	2-3 Prowords explained correctly	>= 4 Prowords explained correctly with examples	All 5 explained fully and correctly with real-world examples
[S2] NATO Phonetic pronunciation	**5**	>= 3 letters or numbers mispronounced	60-79% correct pronunciation	>= 80% correct; errors do not impede understanding	Fully accurate; natural rhythm and pace
[A] Reflex for handling misunderstanding	**3**	Does not handle or handles incorrectly	Handles but slowly and without confidence	Handles correctly and confidently without	Immediate and professional; prevents

Assessment Criteria	Max Score	1 - Below Standard	2 - Minimum	3 - Meets Standard	4 - Outstanding
		when crew asks to repeat		excessive delay	misunderstanding from escalating
	20	< 60% = Below Standard		60-79% = Meets Standard	>= 80% = Outstanding

APPENDIX: GUIDE TO USING LLMs FOR SECONDARY SCENARIO GENERATION

This section provides principles, a master prompt template, and a chapter-by-chapter prompt summary for VJAA instructors to use Large Language Models (Claude, GPT-4, Gemini) to generate additional secondary scenario exercises from the sample scenarios in this document.

A. 5 Principles for Using LLMs in Aviation Training

- LLMs are SUPPORT TOOLS - Instructors MUST always review and adjust outputs before using them in teaching.
- Provide FULL CONTEXT - Course name, reference document (VJC-VJAA-M-125), student level, learning objectives.
- Reference SPECIFIC STANDARDS - Cite the exact section in the source document - do NOT let LLMs invent procedures.
- Specify OUTPUT FORMAT - Dialogue table, 4-level rubric, bullet points, comparison table.
- VERIFY ACCURACY - LLMs may generate incorrect terminology or procedures inconsistent with VJC. ALWAYS cross-check against the source document.

B. Master Prompt Template

SECONDARY SCENARIO GENERATION PROMPT (LLMs: Claude / GPT-4 / Gemini)

```
[ROLE] You are an [aviation training expert / instructor / safety specialist]
at VJAA (Vietjet Aviation Academy).  [DOCUMENT CONTEXT] Reference: VJC-VJAA-M-
125 "Communication with Flight Crew" 5 chapters: Ch.1 Pushback | Ch.2
Communication Forms | Ch.3 Procedures | Ch.4 Safety | Ch.5 Signals  [BASE
SCENARIO] [PASTE THE SAMPLE SCENARIO CONTENT TO BE EXPANDED HERE]  [TASK:
CREATE SECONDARY SCENARIOS] Based on the above, create [NUMBER] secondary
scenarios with the following changes: - Variable changes: [aircraft type /
airport / time / tug type / weather / etc.] - Added complexity: [APU INOP /
headset failure / bad weather / pax on board / etc.] - Difficulty level:
[Basic / Intermediate / Advanced] - Exercise type: [Role-play / MCQ / Written
/ Fill-in-the-blank]  [OUTPUT FORMAT] 1. Scenario information table (6 rows:
Aircraft / Location / Equipment / Conditions / Reference / Objective) 2.
Student requirements (3-5 bullets) 3. Dialogue script (2-column table: Speaker
| ICAO-standard content) 4. Suggested answer with explanation 5. Scoring
rubric: 5 criteria x 4 levels (Below Standard / Minimum / Meets Standard /
Outstanding) + total score 6. 2-3 open discussion questions (optional)
[CONSTRAINTS] - Use ICAO standard English for all radio calls - Do not invent
procedures not present in VJC-VJAA-M-125 - Length per scenario: 300-500 words
```

C. Chapter-by-Chapter Prompt Summary

Chapter	Suggested Prompt Keywords	Changeable Variables	Additional Complexity Factors
Ch.1 Pushback & Towing	push-back, towing, towbar, towbarless, hand signals, towbar failure	Aircraft type (A320/A321/A330/COMAC), airport, push direction, tug type, sequence	Pin shear, APU INOP, headset fail, nighttime, engine fire during pushback

Chapter	Suggested Prompt Keywords	Changeable Variables	Additional Complexity Factors
Ch.2 Comm Forms	arrival, departure, GPU, ACU, ASU, COMAC, BEACON, headset	A320/A321/A330/COMAC C909, arrival/departure, airport (SGN/HAN/DAD/PQC)	APU INOP, crew not responding, fog, pax on board + refueling
Ch.3 Procedures	walk-around, refueling, turnaround, HI LVL, 800VU, CRS, GO	Aircraft type (A320/A330), airport, refueling conditions, number of abnormalities	HI LVL activates early, CRS busy, pax boarding, nighttime, multiple simultaneous abnormalities
Ch.4 Safety	SHUTDOWN, GSE accident, FOD, thunderstorm, fire, lightning, collision	Number of simultaneous tasks, incident type, timing, airport, urgency level	4+ tasks during SHUTDOWN, pushback in motion, nighttime, headset unavailable
Ch.5 Signals	marshalling, NATO phonetic, prowords, wands, hand signals	Aircraft type, nighttime, visibility conditions, proword type, callsign	Wand light failure, limited visibility, escalating misunderstanding, wrong proword consequences

--- END OF DOCUMENT ---

Sample Scenario Exercise System | Communication with Flight Crew | VJC-VJAA-M-125
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